Ishan Mishra

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Research Interests: Planetary Science & Astronomy, Spectroscopy and Photometry of Planetary Surfaces in the Solar System and Beyond

Career & Education

- Staff Scientist — Dec 2025 -
  IPAC, California Institute of Technology
- NASA Postdoctoral Program Fellow (JWST Project) — Nov 2025 - Dec 2025
  NASA Goddard Space Flight Center, USA
- JPL Postdoctoral Fellow — Oct 2022 - Oct 2025
  NASA Jet Propulsion Laboratory, California Institute of Technology, USA
- PhD in Astronomy — Jul 2017 - Aug 2022
  Cornell University, USA
- M.S. in Astronomy — Jul 2017 - Feb 2020
  Cornell University, USA
- B.Tech. in Electronics & Communications Engineering — Jun 2012 - May 2016
  Indian Institute of Technology, Guwahati, India

Honors and Awards

- NASA Postdoctoral Program (NPP) Fellowship (JWST Project) — 2025
- OPAG Meeting Early Career Travel Stipend, *USRA* — 2022
- Graduate Conference Grant, *Cornell University* — 2018, 2021
- Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellowship — 2020
- Research Travel Grant (suspended due to COVID-19), *Cornell University* — 2020
- Carl Sagan Institute Travel Grant, *Cornell University* — 2018,2020
- Other Worlds Lab Summer Research Fellowship, *U.C. Santa Cruz* — 2019
- Summer Research Fellowship, *Academia Sinica Institute of Astronomy and Astrophysics, Taiwan* — 2016
- KVPY fellowship (declined), *Indian Academy of Sciences* — 2011

Peer-Reviewed/Refereed Publications

- DeColibus, R. A., + 5 co-authors including **Mishra, I.**, *Optical Spectroscopy of the Uranian Moons from Equinox to Northern Summer*, 2026, *Planetary Science Journal* (in press).
- Belgacem, I., + 6 co-authors including **Mishra, I.**, *Spectrophotometry of selected regions of Europa – Revisiting Galileo NIMS data*, 2025, *Planetary Science Journal*, 6, 10.
- **Mishra, I.**, Dhingra, R., Buratti, B. J., Seignovert, B., & White, O. L., *Investigating the extent of bladed terrain on Pluto via photometric surface roughness*, 2025, *Journal of Geophysical Research: Planets*, 7, 130.
- Lewis, B. L., + 20 co-authors including **Mishra, I.**, *Improving undergraduate astronomy students' skills with research literature via accessible summaries: An exploratory case study with Astrobites-based reading assignments*, 2025, *Physical Review Physics Education Research*, 21, 010124.
- Scully, J. E. C., + 18 co-authors including **Mishra, I.**, *Potential landing sites: a comprehensive reconnaissance assessment of the Europa Clipper trajectory*, 2025, *Planetary Science Journal*, 6, 142.
- **Mishra, I.**, Lewis, N., Lunine, J., & Hand, K. P., *An assessment of organics detection and characterization on the surface of Europa with infrared spectroscopy*, 2025, *Planetary Science Journal*, 6, 130.
- Buratti, B. J., Pittichova, J., **Mishra, I.**, & 11 co-authors, *Pre-impact albedo map and photometric properties of Dimorphos from DART and ground-based data*, 2024, *Planetary Science Journal*, 5, 83.

- First, E., **Mishra, I.**, & 4 co-authors, *Mid-infrared spectra for basaltic rocky exoplanets*, 2024, *Nature Astronomy*, 1-10.
- Becker, T. M., + 61 co-authors including **Mishra, I.**, *Exploring the composition of Europa with the upcoming Europa Clipper mission*, 2024, *Space Science Reviews*, 220, 5.
- Johnson, A.B., + 21 co-authors including **Mishra, I.**, *A SmallSat mission study for STARLITE: superluminous tomographic atmospheric reconstruction with laser-beacons for imaging terrestrial exoplanets*, 2024, *Proc. SPIE 13092, Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave*, 130922U
- **Mishra, I.**, Lewis, N., Lunine, J., Helfenstein, P., MacDonald, R. J., Filacchione, G., & Ciarniello, M., *Bayesian analysis of Juno/JIRAM's NIR observations of Europa*, 2021, *Icarus*, 357, 114215.
- **Mishra, I.**, Lewis, N., Lunine, J., Hand, K. P., Helfenstein, P., Carlson, R. W., & MacDonald, R. J., *A comprehensive revisit of select Galileo/NIMS observations of Europa*, 2021, *Planetary Science Journal*, 2, 183.
- Lewis, N. K., + 21 co-authors including **Mishra, I.**, *Into the UV: The atmosphere of the hot Jupiter HAT-P-41b revealed*, 2020, *The Astrophysical Journal Letters*, 902, L19.
- Bhattacharya, S., **Mishra, I.**, Vaidya, K., & Chen, W.P., *Disintegration of the Aged Open Cluster Berkeley 17*, 2017, *The Astrophysical Journal*, 847, 138.

Manuscripts Under Review

- MacDonald, R. J., + 22 co-authors including **Mishra, I.**, *The atmosphere of a white dwarf planet*, *Nature.*
- **Mishra, I.**, *FROSTIE: An open-source modeling and retrieval package for spectroscopic data of planetary surfaces*, *Journal of Open Source Software (JOSS).*

Manuscripts in Preparation

- **Mishra, I.**, Buratti, B. J., *Comprehensive analysis of surface roughness across various terrains on Europa*, in prep for *Planetary Science Journal.*

Relevant Conference Proceedings/Abstracts

- MacDonald, R. +21 co-authors including **Mishra, I.**, 2024. *A JWST Transmission Spectrum of a White Dwarf Exoplanet.* AAS/Division for Extreme Solar Systems Abstracts.
- White, J., **Mishra, I.**, Lewis, N. 2021. *High Resolution JWST Transmission Spectra and Their Retrievals for GJ1214b.* American Astronomical Society Meeting Abstracts.
- Lewis, N. +13 co-authors including **Mishra, I.**, 2019. *Unlocking the Hidden Secrets of Hot Jupiter Atmospheres through Near-Ultraviolet Spectroscopy: A Case Study of HAT-P-41b.* AAS/Division for Extreme Solar Systems Abstracts.
- Buratti, B. J., Pittichova, J., **Mishra, I.**, et al., 2023. *Defining the Albedo and Photometric Properties of Dimorphos in Preparation for the Hera Encounter.* AAS/Division for Planetary Sciences Meeting Abstracts #55.
- Belgacem, I., Buratti, B. J., Cornet, T., **Mishra, I.**, 2024. *Is It Roughness or Albedo? Comparing Apples to Apples.* 55th Lunar and Planetary Science Conference 3040.
- **Mishra, I.**, Buratti, B. J. 2023. *Penitentes on Europa: Finding the Photometric Evidence.* AGU Fall Meeting Abstracts.
- **Mishra, I.**, Buratti, B. 2023. *Unique determination of macroscopic roughness of small airless planetary bodies from spacecraft images.* AAS/Division for Planetary Sciences Meeting Abstracts #55.
- Mishra, I., Lewis, N., Lunine, J., Hand, K. P. 2023. *An Assessment of Organics Detection and Characterization on the Surface of Europa with Infrared Spectroscopy.* 1st Workshop on Ices in the Solar System: A Volatile Excursion from Mercury and the Moon to the Kuiper Belt and Beyond 2809.
- **Mishra, I.** 2023. *FROSTIE: An Open Source Modelling and Retrieval Package for Spectroscopic Data of Planetary Surfaces.* 6th Planetary Data Workshop 2991.

Approved Grants

- Europa Clipper Work-MoRE Program (Cycle A) 2025
  An Omnibus Photometric Model for Europa
  (Task Lead: **Mishra, I.**)
- Europa Clipper Work-MoRE Program (Cycle A) 2025

Topographic modeling to simulate multi-scale roughness for Europa Clipper Investigations and Science
(Task Lead: Sutton, S.; Team Member: **Mishra, I.**)

- NASA Postdoctoral Program (NPP) Fellowship (JWST Project) 2025
  Benchmarking and Enhancing Surface Spectroscopy Models with JWST:
  A Study of Space Weathering Across Small Bodies
- NASA Europa Clipper ICONS Internship Program 2025
  Providing a NIMS Context to Recent JWST Observations of Europa
  (Mentor: **Mishra, I.**; Funding includes one month of salary support)
- JWST Cycle 4, GO 9033 2025
  By the Ashes of Stars: A Chemical Census of a White Dwarf Planet
  (PI: MacDonald, R.J. + 23 Co-I's including **Mishra, I.**)
- JWST Cycle 1, GO 2358 2021
  Revealing the Atmospheric Composition of a White Dwarf Planet
  (PI: MacDonald, R.J. + 14 Co-I's including **Mishra, I.**)
- Future Investigators in NASA Earth and Space Science and Technology (FINESST) 2020
  Grant Number 80NSSC20K1381
  Providing new constraints on Europa's surface composition
  (FI: **Mishra, I.**; PI: Lewis, N.K.)

Select Presentations

- *Testing the penitente hypothesis on Europa via photometric roughness* (**talk**)
  European Geosciences Union (EGU) General Assembly, Vienna 2024
- *The utility of photometric roughness in characterizing*
  the physical nature of planetary surfaces (**lightening talk**)
  30th Meeting of the NASA Small Bodies Assessment Group (SBAG), Tucson, Arizona 2024
- *Decoding the chemical and physical nature of airless planetary bodies:*
  examples from studies of Europa and Pluto (**invited talks**)
  Colloquium for Department of Physical and Astronomy, Cal State LA 2023
  Colloquium for School of Earth and Planetary Sciences, NISER, India 2024
- *Europa Clipper: Exploring an Alien Ocean World* (**invited talks**)
  SPACEweek at Dawson College, Montreal 2023
  Planetary Lunch Seminar, Department of Astronomy, Cornell University 2023
- *How Widespread are the Bladed Terrains on Pluto?* (**invited talk**)
  New Horizons Science Team Meeting 2023
- *Providing new constraints on Europa's surface composition* (**invited talks**)
  University of Arizona Origins Seminar Series 2022
  UC Berkeley CIPS Seminar Series 2022
  Yale Exoplanets and Stars Seminar Series 2022
  Carnegie Earth and Planets Laboratory exoplanets seminar 2022

Outreach/Media

- *Research Highlight: "Skyscraper-sized spikes of methane ice on Pluto"* 2025
  Featured in Live Science and other media outlets for work on Plutonian bladed terrain.
- *Panelist for press event 'NASA's New Horizons: Distant Discoveries in the Outer Solar System'* 2023
  54th Lunar and Planetary Science Conference Hands-on learning projects and workshops conducted virtually for school kids to get them interested in STEM
- *Author for* astrobites.org 2020-2023
  Write brief paper summaries accessible to undergraduate level students

Educational Projects

- *Subject Matter Expert, NASA Infiniscope "Are We Alone?" Project* (Arizona State University) 2025–
  Serving as a scientific content expert for the NASA-funded Infiniscope project, collaborating with ASU science educators to design an interactive digital learning module on Europa.

Teaching/Mentoring experience

- *Mentor, NASA Europa Clipper ICONS Internship Program* Summer 2025
  Supervised a summer undergraduate research project titled *"Providing a NIMS Context to Recent JWST Observations of Europa"*.
- *Graduate student mentor for Cornell Astronomy's REU program* Summers 2020 & 2021
  Mentored two undergraduate students in separate projects involving analysis of transmission spectra of super-Earths and reflectance spectra of rocky planetary surfaces respectively.
- *Astronomy Teaching Assistant (ASTRO 2202)* (Cornell University) Fall 2018
  Assisted in teaching students how to write popular science articles, graded written homework and gave feedback and held office hours.
- *Astronomy Teaching Assistant (ASTRO 1101/1102)* (Cornell University) Fall 2017-Spring 2018
  Taught recitation sections and lab sections that involved exercises on introductory astronomy topics. Assisted in writing and grading homework assignments and exams.

Professional development certificates

- *NASA Planetary Science Summer School* 2024
  Served as the project manager and science team member for the mission concept CORA: a New Frontiers class lander for in-situ chemical and geophysical exploration of Ceres' Occator Crater.
- *Cornell Small-sat Mission Design School* 2023
  Served as the instrument lead for the mission concept STARLITE: a constellation of smallsat laser-beacons to enable multi-conjugate adaptive optics (MCAO) for ground-based direct-imaging of exoplanets.

Professional service and affiliations

- Member of NASA SMD Review Panels 2023-
- Member of the Science Organizing Committee (SOC) 2023
  of the joint *DPS-EPSC 2023* meeting
- Board member of the Foreign-national Advocacy Network (FAN), 2023-
  an employee resource group at JPL
- Reviewer for *AAS* Journals and *Icarus* (2 papers reviewed), 2021-
- Graduate student affiliate, *Europa Clipper Mapping and Imaging Spectrometer (MISE) team* 2020-2022
- Postdoctoral Affiliate, *Europa Clipper Project Science team* 2022-
- *Co-organizer of Cornell Astronomy REU Program's Python workshop* 2020
  Conducted a multi-day workshop on introduction to scientific programming with python.
- *Co-organizer of Introduction to GitHub workshop at ERES-V* 2019
  Conducted a 3-hour workshop on open-source software development using GitHub.